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Project Title:	Risk Assessment of Space Radiation-Enhanced Colon Tumorigenesis		
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Task Description:	This proposal addresses the Carcinogenesis category of the Bioastronauts Roadmap by investigating carcinogenic risk of human colonic cells and tissues after low and high LET radiation exposure. The colon is a major target for the carcinogenic effects of radiation. Aberrant crypt foci are an early precursor lesion for colorectal cancer (CRC) that is characterized by trisomy 7 (present in 37% of adenomas and 80% of adenocarcinomas). The appearance of the adenomatous polyp becomes more prevalent with increased age (10% incidence at age 40 and >25% by age 60). Pre-cancerous lesions may already exist or develop in astronauts going on long-term missions, the result of which would be an increased risk for cancer development through exposure to space radiation. Our key objective is to determine the probabilities for increased colonic carcinogenesis from space radiation as a result of radiation dose, quality, and dose rate. We will expose 2D and 3D human colonic cells and human colonic biopsies to energetic protons (H), high Z and energy particles (HZE), and x-rays and analyze results using both biological assays and genome-wide systems approaches. We will compare these results to a novel immortalized human trisomy 7 colonic epithelial cell line and will model carcinogenic risk of these cells compared to normal and immortalized diploid colonic cells. We will examine DNA damage sensing, signaling, and repair kinetics in normal and trisomy 7 human colonic cell lines. Finally, we will determine if a novel oral bioavailable compound CDDO-Me (bardoxolone methyl, RTA 402) is an effective and non toxic countermeasure for protecting normal colonic cells and tissues from the detrimental effects of low and high LET irradiation. This research should improve our mechanistic understanding for the variability of radiation dose, quality, and dose-rate dependencies for colon cancer development and provide biological markers for modeling and risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of methods to maintain human colonic epithelial cells in culture long term has many benefits for life on Earth. The unique human colonocyte reagents we have developed may help in our understanding of the earliest events in the development of human colon cancer. In addition to normal human colonocytes, we have also isolated the first pre-cancerous human colonocyte cell culture that has one extra chromosome (trisomy 7). Trisomy 7 occurs in more than 37% of sporadic colon cancer and is believed to be one of the earliest steps in colon cancer progression. Our goal is to identify the underlying molecular mechanisms leading to early genetic changes in colon cancer and to validate novel genetic alterations that may occur as critical steps in colon cancer progression. We have initiated experiments to silence and express candidate genes known to be altered in colon cancer into our colonocyte cell lines. Experiments conducting genome-wide sequencing of colon tumors revealed that in addition to the common already known gene mutations, many additional genes are mutated in colon cancer. It is not known which of these mutated genes are important for colon cancer progression. Development of an increasingly cancerous phenotype in the colonocytes after specific genetic alterations should help validate the currently known pathways leading to colon cancer and identify new targets for therapeutic interventions.

Risk Assessment of Space Radiation-Enhanced Colon Tumorigenesis
Taskbook Report

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This project investigates the cellular and molecular mechanisms that contribute to the carcinogenic risk of human colonic cells and tissues after occupational radiation exposure. This proposal addresses the Carcinogenesis category of the Bioastronauts Roadmap. Our key objective is to determine the probabilities for increased colonic carcinogenesis from space radiation as a result of radiation dose, quality, and dose rate. Assuming the results of our studies indicate an increase in fatal colon cancer risk exceeds 3% we have also initiated pilot studies on a novel medical countermeasure (CDDO-Me) that is oral available, currently in Phase IIb clinical trials for chronic kidney disease, and is an agent that has minimal toxicities. We will determine in experimental cell culture models and eventually in mouse models if this countermeasure will reduce the probability of increased carcinogenesis.

Human colonic epithelial cells have proven to be exceptionally difficult to propagate in culture. Inappropriate crypt isolation techniques, inadequate cell culture substrates and nutrients, and oxidative damage have possibly contributed to the short in vitro lifespan (about ten days) of an otherwise highly replicative cell type. One of our major accomplishments has been to develop a method of sustained division of normal human colonic epithelial cells. In addition, we have immortalized these cells by introduction of cyclin-dependent kinase 4 (cdk4) and the catalytic component of telomerase (hTERT). These cells have now been in continuous culture for over two years, and we continue to characterize these cells as a human epithelial cell culture model for irradiation studies. We have also developed methods to culture these cells in 3D.

We have already successfully accomplished many of the specific objectives of this project, including survival curves with several doses of HZE charged particle types. Another major accomplishment has been comparing the effects of radiation on colonic epithelial cells in monolayers and in intact human colonic tissues. Immortalized colonic epithelial cells were irradiated at the NASA Space Radiation Laboratory (NSRL) with either 1Gy (1GeV) 56Fe or 1Gy (1GeV) protons and at various time points were stained for DNA damage responses. Results show more persisting damage at 24 hours with 56Fe particle irradiation compared to protons. We compared these results to colonic tissue biopsies obtained from normal human volunteers irradiated with 1 Gy gamma rays or 1Gy (1GeV) 56Fe. Radiation-induced DNA damage and repair observations in low LET gamma irradiated specimens reveal more pronounced early DNA damage responses in the epithelial cell compartment compared to the stromal cell compartment. The low LET damage foci mostly

Task Progress:

disappear at 24 hours. High LET irradiated biopsy specimens show a pattern of DNA damage response predominance in the epithelial cell compartment. Un-repaired DNA damage in colonic epithelial cells and the differing repair responses between the epithelial and mesenchymal compartments in tissues may enhance tumorigenesis by both stem cell transformation and alterations in the radiation-induced permissive tissue microenvironment that may potentiate cancer progression. This work has led to several presentations at NASA workshops and has recently been published (Radiation Research 171:33-40, 2009).

Summary of progress:

1. DNA damage in colonic epithelial cells is more difficult to repair when caused by iron particles compared to protons and gamma rays.

2. Iron damage persists longer in both 2D and 3D colonic epithelial cell cultures compared to protons and gamma rays.

3. Epithelial cells in intact human colonic tissues display more damage foci than surrounding stromal cells after both high and low LET radiation. All cells in intact colonic tissues are exposed to the same radiation dosage and are likely experiencing equal amounts of double strand breaks, but location in the cell cycle may explain why the replicating epithelial cells show more pronounced DNA damage response compared to quiescent stromal cells.

4. Complex DNA damage caused by high LET particles and paucity of DNA damage response in quiescent cells may lead to long standing cellular alterations (cell senescence) that can make the tissue microenvironment more permissive for tumor progression.

5. Additional cells from another patients have been established and characterized. We have an immortalized normal diploid cell line and one that is immortalized with one chromosomal change (trisomy 7). Trisomy 7 is present in about 65% of all colonic adenoma and adenocarcinoas. We have now progressed these cells partially toward tumorigenicity by reducing p53 expression (shRNA to p53), overexpressing K-ras and c- myc and are currently testing if irradiation converts these cells to cancer forming cells.

6. We have also passaged the cells with trisomy 7 (contain hTERT, cdk4, shRNAp53 and mutant K-ras) and observed that a subclone of these cells contain a trisomy 20 (also very common in colon cancer). While these cells make large colonies and grow anchorage independent in soft agar they do not make tumors in nude mice. We have now expressed c-myc in these cells and are currently testing if these cells are tumorigenic. The isogenic partially progressed cell lines are extremely useful for radiation experiments to determine biologically distinct steps in cancer progression.

7. We have now demonstrated the ability to differentiate the immortalized human colonic epithelial cell lines in Matrigel®. Very recently another laboratory reported that single LGR5 positive stem-like cells from mouse small intestine build crypt-villus structures in Matrigel without a mesenchymal niche. (Sato et al Nature, 459:262 2009). We now have obtained results showing that our human colonic epithelial cells express several gastrointestinal putative stem cell markers including CD44, CD29, BMI-1, LGR5 and others. When these cells are placed in Matrigel® with or without a stromal feeder layer we observe morphologically almost identical structures that were reported in the Sato et al manuscript. This indicates that our immortalized human colonic epithelial cells can make similar 3D differentiated structures as do primary cells from mouse small intestine. Our cells are clonal, can divide long-term (are immortalized), and single cells divide to generate self-organizing structures with several differentiated cell types and architecture of crypts present in adults in the absence of a cellular niche. Our cells demonstrate polarized villin staining, junction formation (ZO-1), a subset of cells express Muc2 similarly to the staining of mouse small intestinal cells, and some even express chromogranin A (a neuroendocrine marker). This helps demonstrate that we are culturing human colonic epithelial stem cells.

New Directions: Based on our previous experimental results we have now moved into the following directions to progress our studies.

Specific Aims

Aim 1. To examine for biological changes and genomic signatures after irradiation of normal diploid, immortalized diploid, immortalized trisomy 7, and partially progressed human colonic epithelial cells (HCECs).

Key objectives include:

• To complete the genomic analysis of the isogenic lines of human colonic epithelial cells with and without exposure to low LET and high LET irradiation. We will be using 600 MeV 56Fe instead of 1 GeV and will be testing doses from 0.1 Gy up to 1Gy focusing on NASA chosen priority beam types.

• To complete the survival curves of the colonic epithelial cells and to compare these to those we have already completed with human lung epithelial cells.

• To progress immortalized human colonic epithelial cells toward colon cancer. Partially progressed cells will then be exposed to low and high LET radiation to determine if saturation density, clonogenicity, anchorage independent growth in soft agar, or invasion into matrigel as surrogate markers of stages in tumor progression changes.

Anticipated Results

We should be able to identify gene signatures of a panel of isogenic human colonic cells (normal and preneoplastic) after low and high LET irradiation. These will be compared to clearly defined biological endpoints (e.g. increased growth rates, increases saturation density, increased clonogenicity and colony formation in soft agar). We will focus on gene expression arrays and array CGH. We do not expect any major confounding issues as we now have two years experience with these human cellular reagents.

Aim 2. To determine if a novel oral bioavailable compound CDDO-Me (bardoxolone methyl, RTA-402) is an effective and non toxic countermeasure for protecting normal cell or tissues from the effects of low and high LET irradiation.

Key objectives include

• Determining if CDDO-Me reduces biological endpoints such as saturation density, clonogencity, anchorage independent growth in soft agar, and invasion into matrigel after irradiation with low and high LET radiation in normal and partially progressed human colonic epithelial cells.

	• To test if CDDO-Me mitigates low and high-LET irradiation induced progression Anticipated Results In an ongoing collaboration with Reata Pharmaceutical, Inc. (Irving, TX), we have demonstrated that methyl-2-cyano-3,12-dioxooleana-1,9(11)-dien-28-oate (CDDO-Me, BARD, bardoxolone methly), an antioxidant inflammation modulator that is currently in human clinical trials, protects normal tissues from radiation-associated inflammation by inhibiting NF-kB activity, inducing Nrf2, suppressing many pro-inflammatory cytokines. CDDO-Me, is in Phase IIb clinical trials to determine in patients with chronic kidney disease if it improve renal functions (e.g. serum creatinine, creatinine clearance, uric acid, angiotensin II, cystatin C, etc). As the exposure to highly energetic charged particles, such as galactic cosmic radiation (GCR) and solar proton events, may limit the amount of time astronauts can safely spend in space, it is critical to explore and develop both shielding and dietary/medical countermeasures that may reduce the overall risks of developing fatal cancer after long-term space missions. Thus, this specific aim is to test in our HCECs strains and lines in the presence and absence of CDDO-Me if we can reduce cancer biological endpoints, such as clonogenicity and growth in soft agar. We will also test if CCDO-Me will partially mitigate the damage and downstream biological consequences of irradiation. Pilot exploratory experiments will be conducted during the next review period. Summary The answer to the very important question of whether exposure to HZE particles enhances cancer risk in humans remains elusive, especially in highly proliferative tissues such as the gastrointestinal system where only low LET RBE are determined for human colonic tissue. In this proposal, we have initiated experiments to test the hypothesis that high energy HZE particles generate a different and/or unique radiation response in mammalian cells derived from the colon as compared to gamma-rays. The proposed research has revealed fundamental differences in cellular responses between low and high LET exposures in human normal colonic cells, immortalized colonic cells, and intact human colonic tissue biopsies. These studies should provide gene expression signature along with array SNP analyses that may correlate with biological markers for risk assessment. Obtaining these critically important results should facilitate modeling to further assess risk as well as guide direction for radiation protection for astronauts during space missions in the future. This proposal primarily addresses the category in the Bioastronauts Roadmap involving carcinogenesis that is associated with assessing risk of increased cancer morbidity or mortality in astronauts caused by occupational radiation exposure. Specifically we are addressing molecular and genetic factors that contribute to carcinogenic risk of human colonic cells (a tissue with high cellular turnover and that is particularly sensitive to the effects of radiation).

Bibliography Type:	Description: (Last Updated: 11/21/2025)
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Abstracts for Journals and Proceedings	Eskiocak U, Roig AI, Kitten E., Lulla D, Wright WE, Shay JW. "RTA402 protects against radiation-induced neoplastic transformation of normal human colonic epithelial cells." Presented at the Heavy Ions in Therapy and Space Symposium, 5th International Workshop on Space Radiation Research, Cologne, Germany, July 7-10, 2009. Heavy Ions in Therapy and Space Symposium, 5th International Workshop on Space Radiation Research, Cologne, Germany, July 7-10, 2009. , Jul-2009
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Articles in Peer-reviewed Journals	Roig AI, Wright WE, Shay JW. "Is telomerase a novel target for metastatic colon cancer?" Current Colorectal Cancer Reports, in press, July 2009. Expected publication January 2010. , Jan-2010